

THE HERITAGE SCHOOL
FIRST TERM EXAMINATION, 2026-27
MATHEMATICS · CLASS 12 · PRACTICE PAPER 1

Maximum Marks: 80

Time Allotted: Three Hours

Instructions to Candidates

- You are allowed an additional 15 minutes for only reading the paper.
- You must NOT start writing during reading time.
- The Question Paper consists of **twenty questions** and **four sections**: Sections A, B, C and D. **All questions are compulsory.**
- **Section A** consists of very short answer questions of **1 mark** each.
- **Section B** consists of short answer questions of **2 marks** each.
- **Section C** consists of moderately long answer questions of **3 marks** each.
- **Section D** consists of long answer questions of **5 marks** each.
- Internal choices have been provided in **three questions** in each of Sections B, C and D.
- While attempting Multiple Choice Questions in Section A, you are required to write only ONE option as the answer.
- The intended marks for questions or parts of questions are given in brackets [].
- All workings, including rough work, should be done on the same page as, and adjacent to, the rest of the answer.
- Mathematical tables and graph papers are provided.

Syllabus covered: Relations and Functions; Inverse Trigonometric Functions; Matrices; Determinants; Continuity and Differentiability; Applications of Derivatives; Integrals; Application of Integrals; Linear Programming.

Not examined in this paper (held back to the Pre-Board): Differential Equations, Vectors, Three-dimensional Geometry, Probability.

SECTION A – 20 MARKS

Question 1 [20 × 1]

In sub-parts (i) to (xx), choose the correct option or answer the question as instructed.

- (i) Let $A = \{1, 2, 3\}$ and $R = \{(1,1), (2,2), (3,3), (1,2)\}$ be a relation on A. Then R is:
- (a) reflexive and transitive but not symmetric
 - (b) an equivalence relation
 - (c) symmetric and transitive but not reflexive
 - (d) neither reflexive nor transitive
- (ii) The number of onto functions from the set $\{1, 2, 3\}$ to the set $\{a, b\}$ is:
- (a) 6
 - (b) 8
 - (c) 2
 - (d) 0
- (iii) Find the value of $\sin^{-1}(\sin 2\pi/3)$.
- (iv) Write the principal value of $\tan^{-1}(-\sqrt{3})$.
- (v) If $A =$

$$\begin{bmatrix} 0 & 2 & -3 \\ -2 & 0 & a \\ 3 & -1 & 0 \end{bmatrix}$$

is a skew symmetric matrix, find the value of a.

- (vi) If A is of order 3×2 and B is of order 2×5 , write the order of AB.
- (vii) If A is a square matrix of order 3 with $|A| = 2$, find $|\text{adj } A|$.
- (viii) If A is a square matrix of order 3 with $|A| = -3$, find $|2A|$.

- (ix) The greatest integer function $f(x) = [x]$ is discontinuous at:
- every integer
 - $x = 0$ only
 - no point of $\mathbb{R}$
 - every real number
- (x) Differentiate $\log(\sec x)$ with respect to x .
- (xi) If $y = \sin(\log x)$, find dy/dx .
- (xii) If $x = t^2$ and $y = t^3$, find dy/dx in terms of t .
- (xiii) Find the slope of the normal to the curve $y = x^2 - 3x$ at the point where $x = 2$.
- (xiv) The function $f(x) = x^3 - 6x^2 + 9x + 15$ is strictly decreasing on:
- $(1, 3)$
 - $(-\infty, 1)$
 - $(3, \infty)$
 - nowhere on $\mathbb{R}$
- (xv) Evaluate $\int \sec^2 x \, dx$.
- (xvi) Evaluate $\int_1^e \frac{1}{x} \, dx$.
- (xvii) Evaluate $\int_{-\pi}^{\pi} \sin^3 x \, dx$.
- (xviii) Find the area of the region bounded by $y = x$, the x -axis and the lines $x = 0$ and $x = 2$.
- (xix) The corner points of the feasible region of an LPP are $(0,0)$, $(4,0)$, $(2,4)$ and $(0,5)$. Find the maximum value of $Z = 3x + 2y$.
- (xx) **Assertion (A):** if the feasible region of an LPP is unbounded, the objective function may have no maximum value.
Reason (R): an unbounded feasible region contains points arbitrarily far from the origin.
- Both A and R are true and R is the correct explanation of A
 - Both A and R are true but R is not the correct explanation of A
 - A is true but R is false
 - A is false but R is true

SECTION B - 14 MARKS

Question 2 [2]

Show that the function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = (3x + 5)/2$ is a bijection, and find f^{-1} .

Question 3 [2]

Find the principal value of $\cos^{-1}(\cos(13\pi/6))$.

OR

Prove that $\tan^{-1}1/7 + \tan^{-1}1/3 = \tan^{-1}1/2$.

Question 4 [2]

If $A =$

2	-1
3	4

show that $A^2 - 6A + 11I = O$.

Question 5 [2]

The radius of a circle is increasing at the rate of 0.7 cm/s. Find the rate at which its circumference is increasing.

Question 6 [2]

Find k so that the function

$(x^2 - 25)/(x - 5)$	$x \neq 5$
k	$x = 5$

is continuous at $x = 5$.

OR

If $y = x^x$, find dy/dx .

Question 7 [2]

Evaluate $\int (\log x)/x \, dx$.

Question 8 [2]

Find the maximum value of $Z = 3x + 4y$ subject to $x + y \leq 4$, $x \geq 0$, $y \geq 0$.

OR

Write down the corner points of the feasible region determined by $x + 2y \leq 8$, $x \geq 0$, $y \geq 0$.

SECTION C - 21 MARKS

Question 9 [3]

Solve for x : $\tan^{-1}(2x) + \tan^{-1}(3x) = \pi/4$.

OR

Find the value(s) of x for which

$$\begin{vmatrix} x & 2 \\ 18 & x \end{vmatrix} = \begin{vmatrix} 6 & 2 \\ 18 & 6 \end{vmatrix}$$

Question 10 [3]

Using properties of determinants, prove that

$$\begin{vmatrix} a-b-c & 2a & 2a \\ 2b & b-c-a & 2b \\ 2c & 2c & c-a-b \end{vmatrix} = (a+b+c)^3$$

Question 11 [3]

If $y = (\sin^{-1}x)^2$, prove that $(1-x^2)(d^2y)/(dx^2) - xy/(dx) = 2$.

Question 12 [3]

The volume of a cube is increasing at the rate of $9 \text{ cm}^3/\text{s}$. How fast is its surface area increasing when the length of an edge is 10 cm ?

OR

If $y = \tan^{-1}(1+x)/(1-x)$, find dy/dx .

Question 13 [3]

Evaluate $\int (2x+1)/((x+2)(x-3)) \, dx$.

OR

Evaluate $\int e^x \sin x \, dx$.

Question 14 [3]

Evaluate $\int_0^{\pi/2} (\sin x - \cos x)/(1 + \sin x \cos x) \, dx$.

Question 15 [3]

Solve the following linear programming problem graphically: maximise $Z = 5x + 3y$ subject to $3x + 5y \leq 15$, $5x + 2y \leq 10$, $x \geq 0$, $y \geq 0$.

SECTION D - 25 MARKS

Question 16 [5]

Show that the function $f: \mathbb{R} - \{-4/3\} \rightarrow \mathbb{R} - \{4/3\}$ defined by $f(x) = 4x/(3x+4)$ is one-one and onto, and find f^{-1} .

OR

Let R be the relation on $\mathbb{N} \times \mathbb{N}$ defined by $(a, b) R (c, d)$ if and only if $a + d = b + c$. Show that R is an equivalence relation.

Question 17 [5]

Solve the following system of equations by the matrix method:

$$x + 2y + z = 7, \quad x + 3z = 11, \quad 2x - 3y = 1$$

Question 18 [5]

A window is in the form of a rectangle surmounted by a semicircular opening. The total perimeter of the window is 10 m. Find the dimensions of the window that admit the maximum light (that is, that maximise the area).

OR

Prove that the height of a closed right circular cylinder of given volume and least total surface area is equal to its diameter.

Question 19 [5]

Evaluate $\int (x^2 + 1)/((x - 1)^2(x + 3)) dx$.

Question 20 [5]

Find the area of the smaller region bounded by the circle $x^2 + y^2 = 4$ and the line $x + y = 2$.

OR

Find the area of the region bounded by the parabola $y = x^2$ and the line $y = x + 2$.

ANSWER KEY & MARKING NOTES

Practice Paper 1. **Do not read this section before attempting the paper.** Answers were checked against a computer algebra system; the notes give the decisive step rather than a full solution.

Section A

Q	Answer
1 (i)	(a). Every (a,a) is present, so R is reflexive. (1,2) ∈ R but (2,1) ∉ R, so it is not symmetric. The only chain is (1,2),(2,2) and (1,2) ∈ R, so it is transitive.
1 (ii)	(a) 6. Total functions $2^3 = 8$, minus the 2 constant functions.
1 (iii)	$\pi/3$. Since $2\pi/3 \notin [-\pi/2, \pi/2]$, write $\sin 2\pi/3 = \sin(\pi - 2\pi/3) = \sin \pi/3$.
1 (iv)	$-\pi/3$. The range of $\tan^{-1}$ is $(-\pi/2, \pi/2)$ and $\tan(-\pi/3) = -\sqrt{3}$.
1 (v)	$a = 1$. Skew symmetry needs $a_{23} = -a_{32}$, so $a = -(-1) = 1$.
1 (vi)	3×5 . The inner dimensions (2 and 2) must agree and cancel.
1 (vii)	$ \text{adj } A = A ^{n-1} = 2^2 = 4$.
1 (viii)	$ 2A = 2^3 A = 8 \times (-3) = -24$.
1 (ix)	(a) every integer. At an integer n the left limit is $n - 1$ and the right limit is n.
1 (x)	$\tan x$. $d/dx \log(\sec x) = (\sec x \tan x)/(\sec x) = \tan x$.
1 (xi)	$(\cos(\log x))/x$.
1 (xii)	$3t/2$. $dy/dx = dy/dt \div dx/dt = 3t^2/(2t)$.
1 (xiii)	-1 . $dy/dx = 2x - 3 = 1$ at $x = 2$, so the normal has slope $-1/1 = -1$.
1 (xiv)	(a) (1, 3). $f(x) = 3(x - 1)(x - 3)$, which is negative exactly between the roots.
1 (xv)	$\tan x + C$.
1 (xvi)	1. $[\log x]_1^e = \log e - \log 1 = 1$.
1 (xvii)	0. $\sin^3 x$ is an odd function and the interval is symmetric about 0.
1 (xviii)	2 square units. $\int_0^2 x \, dx = [x^2/2]_0^2 = 2$.
1 (xix)	14, at (2, 4). The four values are 0, 12, 14 and 10.
1 (xx)	(a). Both statements are true, and the second is exactly why the first holds: Z can be pushed up without limit along such directions.

Section B

Q	Answer
2	One-one: $(3x_1 + 5)/2 = (3x_2 + 5)/2$ forces $3x_1 = 3x_2$, so $x_1 = x_2$. Onto: solving $y = (3x + 5)/2$ gives $x = (2y - 5)/3$, which is real for every y. Hence $f^{-1}(y) = (2y - 5)/3$.
3	$\pi/6$. Since $(13\pi)/6 = 2\pi + \pi/6$ and cosine has period 2π , $\cos(13\pi)/6 = \cos \pi/6$, and $\pi/6 \in [0, \pi]$.
3 (OR)	With $x = 1/7$, $y = 1/3$ we have $xy = 1/21 < 1$, so the sum is $\tan^{-1}(x + y)/(1 - xy) = \tan^{-1}(10/21)/(20/21) = \tan^{-1}1/2$.
4	$A^2 = [[1, -6], [18, 13]]$, $6A = [[12, -6], [18, 24]]$ and $11I = [[11, 0], [0, 11]]$. Adding entrywise gives the zero matrix.
5	$C = 2\pi r$, so $dC/dt = 2\pi dr/dt = 2\pi(0.7) = 1.4\pi \text{ cm/s} (\approx 4.40 \text{ cm/s})$.
6	For $x \neq 5$, $(x^2 - 25)/(x - 5) = x + 5$, whose limit as $x \rightarrow 5$ is 10. Continuity requires $k = f(5) = 10$.
6 (OR)	$\log y = x \log x$, so $(1/y)(dy/dx) = \log x + 1$ and $dy/dx = x^x(\log x + 1)$.
7	Put $t = \log x$, so $dt = dx/x$ and the integral becomes $\int t \, dt = t^2/2$. Hence $((\log x)^2)/2 + C$.
8	Corner points (0,0), (4,0), (0,4) give $Z = 0, 12, 16$. Maximum $Z = 16$ at (0, 4).
8 (OR)	(0, 0), (8, 0) and (0, 4).

Section C

Q	Answer
9	$(2x + 3x)/(1 - 6x^2) = \tan^{-1}1$, so $6x^2 + 5x - 1 = 0$, i.e. $(6x - 1)(x + 1) = 0$, giving $x = 1/6$ or $x = -1$. At $x = -1$ the left side is $\tan^{-1}(-2) + \tan^{-1}(-3) = -3\pi/4$, so that root is rejected. $x = 1/6$.
9 (OR)	$x^2 - 36 = 36 - 36 = 0$, so $x^2 = 36$ and $x = \pm 6$.
10	Apply $R_1 \rightarrow R_1 + R_2 + R_3$: every entry of the new first row becomes $a + b + c$. Take that factor out, then use $C_2 \rightarrow C_2 - C_1$ and $C_3 \rightarrow C_3 - C_1$ to reduce the determinant to a triangular form; two further factors of $(a + b + c)$ appear, giving $(a + b + c)^3$.
11	$dy/dx = (2 \sin^{-1}x)/(\sqrt{1 - x^2})$, so $\sqrt{1 - x^2} dy/dx = 2 \sin^{-1}x$. Differentiating again and multiplying by $\sqrt{1 - x^2}$ gives the result; verified symbolically.
12	$V = a^3$ gives $9 = 3a^2 da/dt$, so at $a = 10$, $da/dt = 9/300 = 0.03$ cm/s. $S = 6a^2$ gives $dS/dt = 12a da/dt = 12(10)(0.03) = \mathbf{3.6 \text{ cm}^2/\text{s}}$.
12 (OR)	$\tan^{-1}(1 + x)/(1 - x) = \tan^{-1}1 + \tan^{-1}x = \pi/4 + \tan^{-1}x$, so $dy/dx = 1/(1 + x^2)$.
13	Partial fractions give $(3/5)/(x + 2) + (7/5)/(x - 3)$, so the integral is $3/5 \log x + 2 + 7/5 \log x - 3 + C$.
13 (OR)	Applying integration by parts twice and solving for I gives $e^x/2(\sin x - \cos x) + C$.
14	0. Replacing x by $\pi/2 - x$ turns the numerator into $\cos x - \sin x$ while leaving the denominator unchanged, so $I = -I$ and hence $I = 0$.
15	Corner points $(0,0)$, $(2,0)$, $(0,3)$ and $(20/19, 45/19)$. Z there is 0, 10, 9 and $235/19$. Maximum $Z = 235/19 (\approx 12.37)$ at $(20/19, 45/19)$.

Section D

Q	Answer
16	One-one: $(4x_1)/(3x_1 + 4) = (4x_2)/(3x_2 + 4)$ cross-multiplies to $12x_1x_2 + 16x_1 = 12x_1x_2 + 16x_2$, so $x_1 = x_2$. Onto: $y = 4x/(3x + 4)$ gives $x(3y - 4) = -4y$, i.e. $x = 4y/(4 - 3y)$, defined for all $y \neq 4/3$. $f^{-1}(y) = 4y/(4 - 3y)$.
16 (OR)	Reflexive: $a + b = b + a$. Symmetric: $a + d = b + c$ gives $c + b = d + a$. Transitive: from $a + d = b + c$ and $c + f = d + e$, adding gives $a + f = b + e$.
17	With $A = [[1,2,1],[1,0,3],[2,-3,0]]$, $ A = 18 \neq 0$, so A^{-1} exists and $X = A^{-1}B$ gives $x = 2, y = 1, z = 3$. Check: $2 + 2 + 3 = 7$, $2 + 9 = 11$, $4 - 3 = 1$.
18	Let the semicircle have radius r , so the rectangle is $2r$ wide and h tall. Perimeter: $2r + 2h + \pi r = 10$. Area $A = 2rh + 1/2\pi r^2 = 10r - 2r^2 - (\pi r^2)/2$. $dA/dr = 10 - 4r - \pi r = 0$ gives $r = 10/(\pi + 4)$, and then $h = r$. Radius = height = $10/(\pi + 4)$ m ≈ 1.40 m; width = $20/(\pi + 4)$ m ≈ 2.80 m; maximum area = $50/(\pi + 4)$ m² ≈ 7.00 m².
18 (OR)	With $V = \pi r^2 h$ fixed, $S = 2\pi r^2 + 2\pi r h = 2\pi r^2 + 2V/r$. $dS/dr = 4\pi r - 2V/r^2 = 0$ gives $r^3 = V/2\pi$, so $h = V/(\pi r^2) = 2r$. Hence h equals the diameter.
19	Partial fractions: $(3/8)/(x - 1) + (1/2)/((x - 1)^2) + (5/8)/(x + 3)$. Integrating, $3/8 \log x - 1 - 1/(2(x - 1)) + 5/8 \log x + 3 + C$.
20	The line cuts the circle at $(2,0)$ and $(0,2)$. The required area is the quarter disc minus the triangle: $\pi/4(2)^2 - 1/2(2)(2) = \pi - 2$. Area = $(\pi - 2)$ square units ≈ 1.14 square units.
20 (OR)	$x^2 = x + 2$ gives $x = -1$ and $x = 2$, with the line above the parabola between them. Area = $\int_{-1}^2 (x + 2 - x^2) dx = \mathbf{9/2 \text{ square units}}$.